

Mission Plan Interface Volume

For

A-GRA

5.0a

21 APR 2026

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Revision History

Version	Description	Generation Date
5.0a	Initial Public ASK 5.0 Release	21 APR 2026

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1 MISSION PLAN INTERFACE

The L1 Mission Plan (MP) interface defines the Mission Data Package, which Mission Autonomy (MA) must ingest and internally manage to facilitate mission execution. This data includes traditional mission planning inputs, such as weather, airspace boundaries, and airfield data; as well as data specific to MA, including route and task plans. The Mission Data Package should be filled out thoroughly pre-mission, but it can also be updated during mission execution. The MP MMS will be filled out with UCI messages and MA_ extensions which define its data in a future version of the ASK. References to Mission Data Package elements are found throughout the A-GRA, which create dependencies within other interface interactions.

1.1 Interface Description

Mission Data refers to data elements that MA uses to determine what mission it must accomplish, including all constraints, restraints, priorities, and other parameters that may directly or indirectly impact behavior. It is possible for MA to accomplish the vast majority of an end-to-end mission with mission planned data alone, requiring only minimal human-in-the-loop interactions. Mission data is dynamic, allowing C2, Peer, Mission Systems, and contingency driven updates during mission execution.

The supported types of data included in the Mission Data Structure (see figure below) include weather, airspace restrictions, MA configurations, tasking, and Mission Context, etc. Some data is access-restricted, based on Approval Policy data and Authorization to access and modify elements to prevent unauthorized changes during the mission.

Mission Plans are the subset of the Mission Data that provide commands and tasking for MA. They encapsulate Mission Context and constraints to provide MA with the parameters necessary to execute a mission. Not all Mission Data is considered part of the Mission Plan. Some elements such as weather and vehicle characteristics are intended to be independent of any individual Mission Plan and are meant to provide information to MA for the entirety of the mission.

The Mission Plan schema used by MA is based on UCI 2.3, with currently no extensions or custom messages. Existing Mission Plan and related structures within UCI provide a suitable and standardized basis for MA applications. They include an assortment of needed *Plan formats that help to cover many of the facets expected in execution of a Mission. These Sub Plans can include TaskPlan, RoutePlan, and other plans as defined in UCI 2.3. Subordinate plans provide a mechanism to create a hierarchy of Mission Plans with traceability to the originating plan. One or more Mission Plans may be included in the Mission Data Load with Mission Data elements potentially shared between Mission Plans. The Mission Data Load is the collection of all relevant data collected and produced during the pre-mission planning phase. It consists of various routes, plans, and datasets needed to effectively execute the mission. A Mission Plan is assumed to be inactive unless the Mission Data specifically declares its activation. When a Mission Plan is inactive, it is stored for future use and may be used for computational purposes such as state estimation and planning, but it is not considered as a direct source of behavioral guidance. When a Mission Plan is activated, MA is required to attempt to execute its parts to completion in order, which requires tracking progress, which is also used when responding to ad-hoc Direct C2 or contingencies, and resuming the mission plan.

Aspects of the Mission Data are platform and/or implementation specific and not specified in the A-GRA. What is highlighted here is that Vehicles and Mission Systems will both need to interact with the Mission Data so there's a need for majority of the data in a central, accessible location. A-GRA is agnostic to both the process to produce the required data and the storage method. It is assumed a combination of tools such as Joint Mission Planning System (JMPS) will be integrated to produce the entire mission data package.

The format of Mission Data entering MA will conform to the format and requirements outlined in this section. Once deserialized and loaded, the Mission Data is stored and acted upon by MA. Some Mission Data elements are not immediately actionable; they may be schedule-dependent or location-dependent or they may require explicit activation.

MA may receive Mission Data elements from each of the external interfaces. The initial Mission Data defined in the Mission Plan Interface section of the A-GRA provides the baseline Mission Data (including Mission Plans) for MA to execute. Portions of the Mission Data can be modified, updated, and augmented dynamically during mission execution from sources both internal and external to MA via entry points such as the C2 (see Section 4.2), Peer interfaces (see Section 4.3), Mission Systems (see Section 4.6), and Vehicle Interface (see Section 4.7). Mission Data can also be queried and shared between connected entities via the C2 and Peer interfaces. Details of these interactions are captured in their respective sections in the A-GRA. It is important to note that not all elements of the Mission Plan are exposed to each interface, with most operations occurring on a per element basis.

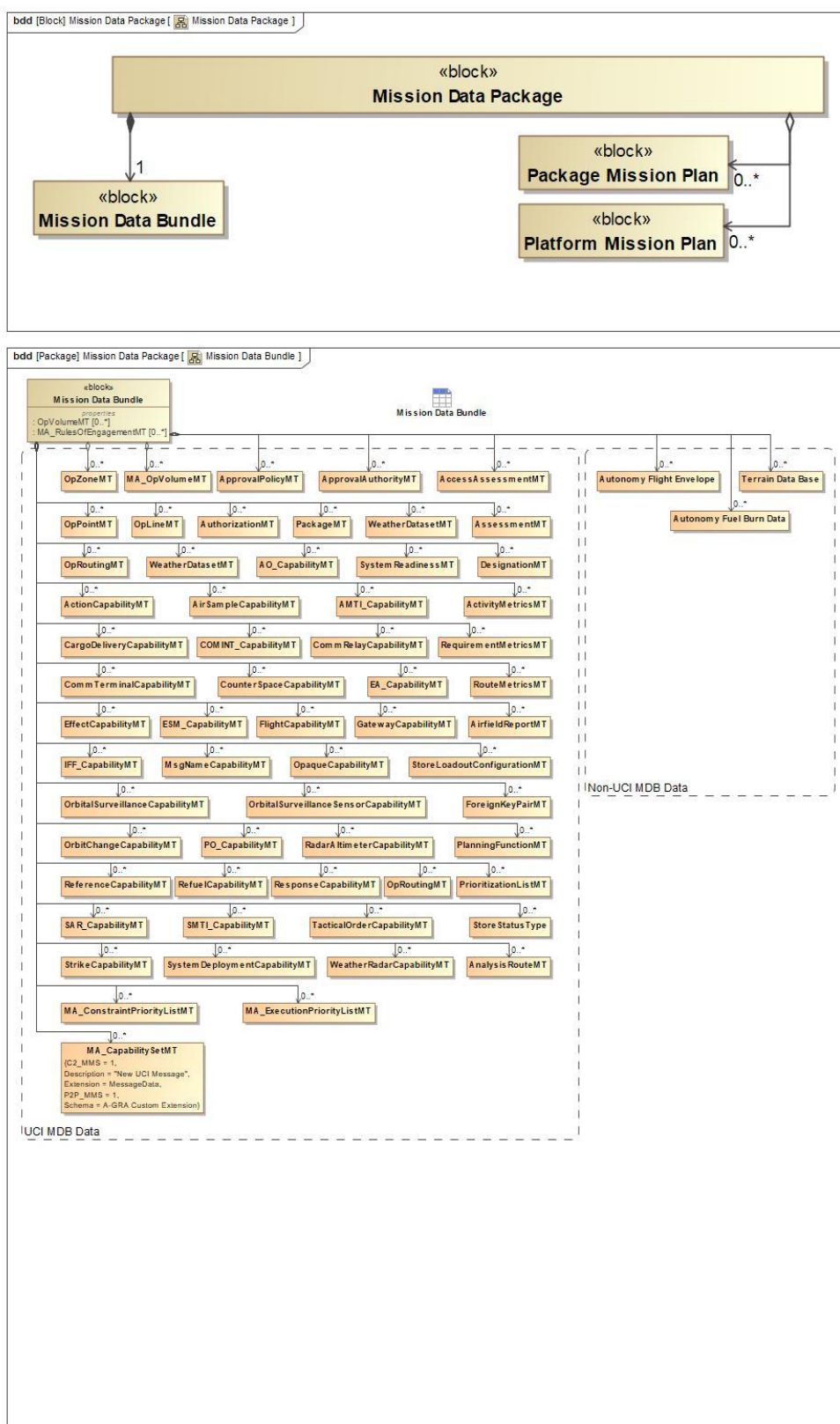


Figure 1-1: Mission Data Package

1.1.1 Mission Plan Message

The Mission Plan schema used by MA is based on UCI 2.3. Existing Mission Plan and related structures within UCI provide a suitable and standardized basis for MA applications. The Mission Plan schema includes various *Plan formats that help to cover many of the facets expected in execution of a Mission. These can include Task Plan, Route Plan, and other plans as defined in UCI 2.3.

Subordinate plans provide a mechanism to create a hierarchy of Mission Plans with traceability to the originating plan.

One or more Mission Plans may be included in the Mission Data Package with Mission Data elements potentially shared between Mission Plans. The Mission Data Package is the collection of all relevant data collected and produced during the pre-mission planning phase. It consists of various routes, plans, and datasets needed to effectively execute the mission. A Mission Plan is assumed to be inactive unless the Mission Data specifically declares its activation. When a Mission Plan is inactive, it is stored for future use and may be used for computational purposes such as state estimation and planning, but it is not considered as a direct source of behavioral guidance. When a Mission Plan is activated, MA is required to attempt to execute its parts.

At the start of each mission or deployment, MA receives the Mission Data Package that will contain the Mission Data needed to execute its mission. The Mission Data Package is used to initialize the Mission Data. Aspects of the Mission Data Package are platform-dependent and implementation-specific and not specified in this ICD. What is highlighted here is that Vehicles and Mission Systems will both need to interact with the Mission Data so there's a need for much of the data to be accessible to these systems. A-GRA assumes that a process exists to generate the Mission Data Package from an existing mission planning tool such as Joint Mission Planning System (JMPS) by augmenting the output from mission planning with additional data needed to support autonomy execution.

1.1.2 Mission Data Bundle

The Mission Data Bundle is a collection of data that contains UCI and non-UCI information that the ACP will use to complete the mission. The Mission Data Bundle will be loaded onto the ACP before the start of the mission, and is required to be present for the mission to begin.

FA will have its own Data Bundle that contains the information that FA will use to operate. It is assumed that any information in FA's Data Bundle that MA will need to access will be contained within the Mission Data Bundle.

Another component of the Mission Data Bundle is the *MA_CapabilitySet* message, which acts as the Minimum Essential Capabilities List (MECL). The purpose of the MECL is to explicitly identify all required Capabilities for the vehicle to successfully complete a given Mission/Mission Plan.

In the *MA_CapabilitySet's MessageData* field, Capability IDs are tied to their respective Applicable System, Mission and/or Mission Plan. The intent of this message is to convey the minimum essential capabilities required to complete a mission/mission plan, which can be evaluated against to detect Mission Capability Fault contingencies. To evaluate MECL, MA sweeps through all listed Capabilities in the *MA_CapabilitySet* and compares them to relevant *CapabilityStatus* messages to ensure the full criteria set is met.

Note 1: The linked Capability messages will contain actual details about capability implementation.

Note 2: Referenced Capabilities should be published ahead of reference.

1.2 Interface Interactions

The L1 MP Interface represents the influence that an external entity, i.e. the Mission Planner, has on MA. It specifies minimum mission plan requirements (i.e. format, data types, fields) for what is expected from mission planners to support MA. These message exchanges are defined and owned by MP, but may be used by other L1 subsystems in their own interfaces with MA, such as C2 and P2P.

1.3 Interface Compliance

The Interface Compliance table contains the compliance requirements for the L1 MP interface.

Table 1-1: Mission Plan Interface Compliance Requirements

ID	Requirement
MA-L1-005	A Level 1 Mission Plan interface shall support the Mission Plan Interface Minimum Message Set defined in the Mission Plan MMS Table in the Mission Plan feature profile (A-GRA location: 1 Problem Domain > 3 Exchange Items > Minimum Message Set (MMS)).

1.3.1 MP MMS

The MP L1 represents the influence that an external entity, i.e. the Mission Planner, has on MA. It specifies minimum mission plan requirements (i.e. format, data types, fields) for what is expected from mission planners to support MA. While the MP L1 isn't an active interface for message exchanges, it does identify what data is expected and who is responsible for providing

1.3.2 MP UCI Extensions

None